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Transcriptional and Post transcriptional Gene Regulation by Non-Coding RNA

Abstract:

Recent advances in genomics have revealed that a significant portion of the human genome is transcribed into non-coding RNAs, particularly long non-coding RNAs (lncRNAs), which play critical roles in gene regulation beyond protein-coding potential. These lncRNAs are now recognized as dynamic regulators operating at both transcriptional and post-transcriptional levels.

This review aims to synthesize current insights into the dual regulatory roles of lncRNAs in controlling gene expression. It focuses on how lncRNAs function in cis and trans to modulate chromatin architecture, transcription factor recruitment, and enhancer–promoter interactions. Additionally, it explores their post-transcriptional roles in splicing, mRNA stability, and translation, including their interactions with microRNAs as precursors or sponges under the competing endogenous RNA (ceRNA) model.

Key themes include well-characterized mechanisms such as Xist-mediated X-chromosome inactivation and HOTAIR-driven trans repression, along with novel findings like enhancer-like lncRNAs and exon–intron circular RNAs (EIcircRNAs). The article also addresses contradictions in lncRNA-protein interactions, particularly the debated role of PRC2 recruitment, highlighting the complexity of RNA-based gene regulation.

lncRNAs represent a highly versatile and context-dependent layer of gene regulation with implications in development, disease, and epigenetic control. While functional diversity is increasingly appreciated, significant gaps remain in understanding their specificity, evolutionary conservation, and therapeutic potential. Future research must focus on unraveling these complexities using integrative and high-resolution tools to fully leverage lncRNAs as molecular regulators and clinical targets.

Key Words: long non-coding RNA(lncRNA), Transcriptional regulation, Post transcriptional regulation, Chromatin remodeling, microRNA(miRNA), Gene expression control.

1. **Introduction**:

In recent years, our understanding of the genome has expanded significantly, revealing that much more of it is transcribed into RNA than previously believed. This shift in perspective has drawn attention to a class of RNA molecules known as **long non-coding RNAs(lncRNAs)-**transcripts that do not code for proteins but appear to play important roles in regulating gene activity. Rather than being biological byproducts, these molecules are now seen as active participants in controlling various aspects of gene expression.

Their wide-spread presence across different tissues and developmental stages suggests that lncRNAs are key regulatory elements, contributing to the fine-tuning of genetic programs in complex organisms. The biological significance of lncRNAs has become increasingly apparent with the discovery of their involvement in a variety of gene regulatory networks. Far from being transcriptional noise, lncRNAs are now understood to be integral regulators of gene expression, functioning through both transcriptional and post-transcriptional mechanisms.

Their ability to interact with DNA, RNA, and proteins allows them to shape the epigenetic and transcriptomic landscapes of the cell. These interactions enable lncRNAs to act as molecular scaffolds, guides, decoys, and enhancers, depending on cellular context and molecular partners.

At the transcriptional level, lncRNAs influence gene expression by mediating chromatin remodeling, facilitating chromatin looping, and modulating the activity of transcription factors. They can bridge DNA and regulatory proteins by serving as structural scaffolds that help recruit chromatin-modifying complexes to specific genomic loci. This scaffolding function can enhance or repress transcription by altering the epigenetic state of chromatin or by promoting the physical proximity of enhancers and promoters.

At the post-transcriptional level, lncRNAs regulate a wide range of processes including splicing, mRNA stabilization, translation, and RNA interference pathways. Many lncRNAs have been found to interact with microRNAs (miRNAs), either by serving as precursors for miRNAs or by acting as competitive endogenous RNAs (ceRNAs) that sequester miRNAs and prevent them from degrading target mRNAs. Through such interactions, lncRNAs contribute to the fine-tuning of gene expression and cellular homeostasis.

The scope of this review is to explore and synthesize current knowledge regarding the dual regulatory role of lncRNAs in transcriptional and post-transcriptional control. And the objective of this review is to provide an integrated overview of how lncRNAs serve as regulatory nodes in gene expression. It aims to highlight the dynamic nature of lncRNA function, the molecular versatility they exhibit, and their potential implications in development, physiology, and disease. By identifying emerging patterns and unresolved questions, this review also seeks to guide future research directions in the field of non-coding RNA biology.

2. What are lncRNAs and why their location matter:

Long non-coding RNA (lncRNA) transcripts longer than 200 nucleotides that don't code for proteins, but they're far from functionless. In fact, their ability to regulate gene expression makes them incredibly versatile players in cell biology.

One key insight is that where lncRNAs are located in the cell often hints at what they do. For instance, Quinn and Chang (2016) found that nuclear lncRNAs are typically involved in controlling gene activity at the DNA level, such as turning genes on or off while cytoplasmic lncRNAs tend to regulate how messenger RNAs are processed, translated, or degraded.

Earlier, Ponting et al. (2009) noted the variety of genomic positions from which lncRNAs arise in which some are nestled between genes, others overlap protein-coding genes. This location affects how they interact with the genome and other molecules.

However, there's debate about how conserved these molecules really are across species. While Ponting's group suggested many lncRNAs might be evolutionarily insignificant due to their low sequence conservation, others like Ulitsky (2016) argued that lncRNAs may retain their function through conserved structure or timing of expression, even if the actual nucleotide sequence varies. This reveals a gap in our current understanding: we still need better ways to predict function beyond just looking at sequences.

3. Function of long non-coding RNA:

3.1. Function of lncRNA in cis

As an initial framework for understanding lncRNA function, lncRNAs can be broadly classified into those that act in cis, influencing the expression or, chromatin state of nearby genes, and those that execute an array of functions throughout the cell in trans. This classification is widely supported by genomic studies, including Ulitsky et al. (2011), who demonstrated cis effects of lncRNAs on neighboring developmental genes, and by Guttman et al. (2011), who described trans-acting lncRNAs modulating distant gene expression networks.

For cis-acting lncRNAs, it is particularly challenging to distinguish functions of the RNA molecule itself from the DNA from which it is transcribed. Engreitz et al. (2016) addressed this by using CRISPR interference to distinguish RNA-dependent from DNA-dependent regulatory effects at lncRNA loci, highlighting the difficulty in parsing these contributions.

3.1.1. Sequence dependent lncRNA regulation in cis

Among the best-characterized cis-acting lncRNAs is Xist, which mediates X chromosome inactivation in female mammals to achieve dosage compensation. Expressed from the X chromosome destined to become inactive (Xi), Xist spreads along Xi, relocating it to the nuclear periphery and promoting the deposition of repressive chromatin marks.

One of the best-characterized cis-acting long non-coding RNAs (lncRNAs) is Xist, which initiates X chromosome inactivation (XCI) in female mammals for dosage compensation. During early embryogenesis, Xist is transcribed from one X chromosome, which becomes the inactive X (Xi). Xist RNA spreads across the Xi, guiding it to the nuclear periphery and initiating transcriptional silencing through deposition of repressive chromatin marks.

The 17-kb Xist transcript contains six repeat domains (A–F), with the A-repeat crucial for silencing. Genetic studies have helped map functional domains of Xist. However, earlier biochemical analyses using standard RIP assays yielded many false positives (Mili and Steitz, 2004). Recent advances, such as crosslinking-based purification and forward genetic screening have enhanced the accuracy of mapping Xist's protein interactome (Chu et al., 2015; McHugh et al., 2015).

A key mechanism involves Xist's A-repeat recruiting SHARP/SPEN, which brings in SMRT and HDAC3, promoting histone deacetylation, an early event in XCI. While whether PRC2 is directly recruited by Xist remains debated, over 25 years of research underscores both progress and complexity in understanding lncRNA-mediated chromatin regulation.

3.1.2. Transcription dependent regulation in cis by lncRNA Loci

Transcription of long non-coding RNA (lncRNA) loci can regulate nearby gene expression in cis, independent of the mature RNA sequence. A classic example is Airn (antisense Igfr2 RNA non-coding), which silences the paternal Igf2r allele via transcriptional interference. This effect is not mediated by the Airn transcript itself but by its antisense transcription through the Igf2r promoter (Latos et al., 2012). Similarly, Engreitz et al. (2016) demonstrated that transcription and splicing of the lncRNA Blustr (linc1319) is crucial for activation of the neighboring gene Sfmbt2. Polyadenylation-induced premature termination of Blustr reduced Sfmbt2 expression and altered chromatin marks (decreased H3K4me3, increased H3K27me3), showing that the transcriptional process itself is essential for licensing gene expression, rather than the resulting RNA.

Anderson et al. (2016) offered a comparable mechanism in the case of Upperhand (Uph), a lncRNA required for proper expression of the heart development gene Hand2. Premature transcription termination of Uph impaired Hand2 expression, enhancer chromatin signatures, and TF binding, though knockdown of the mature RNA had no effect. This suggests a regulatory role for transcription or the nascent transcript, rather than the mature RNA product.

These studies highlight a shared theme: RNA transcription through regulatory regions modulates chromatin state and transcription factor (TF) accessibility, rather than depending on RNA sequence. While mechanisms vary from transcriptional interference to chromatin licensing, they underscore the importance of transcriptional activity itself in cis-regulatory functions of lncRNA loci.

3.2. Functions of lncRNAs in trans

In addition to those that regulate gene expression and chromatin states in cis, there is an increasing number of examples of lncRNAs that leave the site of transcription and operate in trans.

lncRNAs that regulate chromatin states and gene expression at regions distant from their transcription site, such as HOTAIR, which represses HOXD cluster genes by recruiting chromatin-modifying complexes (Rinn et al., 2007), although later studies suggest this repression can occur independently of PRC2 (Portoso et al., 2017).

3.2.1. Gene regulation by lncRNA in trans

HOTAIR, a ~2.2 kb spliced and polyadenylated lncRNA expressed from the HOXC locus, was among the first lncRNAs shown to regulate genes in trans. Unlike the traditional cis-acting lncRNAs, HOTAIR targets the distant HOXD locus to maintain transcriptional repression (Rinn et al., 2007). Its knockdown resulted in upregulation of HOXD genes and reduced H3K27me3, a repressive chromatin mark.

This led to a model in which HOTAIR acts as a scaffold, recruiting chromatin modifiers such as PRC2 to HOXD to establish repression. Chromatin isolation by RNA purification (ChIRP) confirmed HOTAIR's localization to HOXD (Chu et al., 2011), and RIP and RNA pull-down assays revealed associations between HOTAIR and PRC2.

However, subsequent work questioned the specificity of these interactions.
For instance, PRC2 is now known to bind RNA promiscuously in vitro, leading to possible false positives (Davidovich et al., 2015; Mili and Steitz, 2004). Adding to this complexity, Portoso et al. (2017) found that overexpression of HOTAIR in breast cancer cells repressed genes independently of PRC2. Tethering experiments confirmed that HOTAIR could silence a reporter gene without PRC2 recruitment, challenging the earlier assumption that PRC2 recruitment is the primary mechanism of lncRNA-mediated repression.

These findings contrast earlier studies like Zhao et al. (2008), which proposed similar chromatin-recruiting mechanisms for Xist. This contradiction is significant because it challenges the assumption that lncRNA-mediated gene silencing universally depends on PRC2, prompting a reevaluation of RNA-protein interaction specificity in epigenetic regulation. Collectively, evolving methods and newer data emphasize the need for caution in interpreting RNA-protein interactions and suggest that trans-acting lncRNAs like HOTAIR may function via multiple, possibly PRC2-independent, pathways.

4. Post transcriptional regulation by lncRNA

4.1 lncRNA as a source of miRNA

Long non-coding RNAs (lncRNAs) play important roles in post-transcriptional gene regulation, particularly by serving as precursors to microRNAs (miRNAs). miRNAs are derived from primary transcripts (pri-miRNAs) through a two-step processing mechanism. In the nucleus, Drosha and DGCR8 cleave the pri-miRNA into a ~60-nucleotide precursor (pre-miRNA). This pre-miRNA is exported to the cytoplasm and further processed by Dicer and TAR RNA-binding protein (TRBP) into a mature miRNA of 20–23 nucleotides (Winter et al., 2009). Most pri-miRNAs exceed 1 kilobase in length, and are thus often categorized as lncRNAs. Pri-miRNAs may arise from two genomic contexts: they can be embedded within other genes, typically within introns, or they can be transcribed independently from intergenic regions.

Approximately 50% of all miRNAs are produced from non-coding regions, with many located within lncRNAs. This genomic organization suggests that lncRNAs may act not only as passive precursors but may also have regulatory roles themselves, encoded through their exons or untranslated regions. One key example is the lncRNA DLEU2, which harbours the tumor-suppressive miR-15a/16.1 cluster within its third intron (calin et al., 2002). In adult chronic lymphocytic leukemia (CLL), the expression of these miRNAs is controlled by the DLEU2 promoter, under the regulation of transcription factors MYC (Lerner et al., 2009) and PAX5.

However, in childhood acute myeloid leukemia (AML), methylation studies have shown that the miRNA cluster is regulated independently of its host gene, indicating an epigenetic mechanism. Similarly, in breast cancer, miR-31 is embedded in the lncRNA LOC554202 and is downregulated due to promoter methylation of the host gene (Augoff et al., 2012). These comparative examples highlight how lncRNA-hosted miRNAs are regulated differently depending on tissue type, developmental stage, and disease context, reflecting both transcriptional co-regulation and epigenetic decoupling.

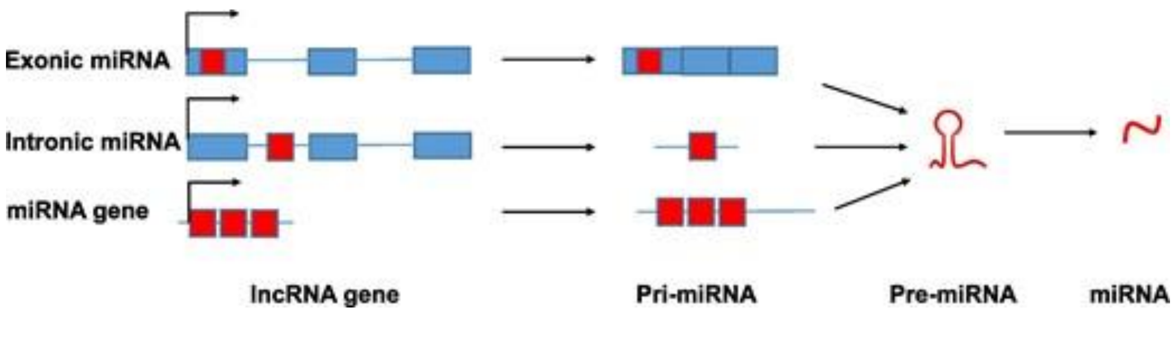

Figure1: lncRNA as a source of miRNA: miRNA sequences can be embedded in lncRNA exons (blue) or introns (line), or arise from independent transcriptional units. Despite different origins, they converge at the pre-miRNA stage. lncRNA: long non-coding RNA; miRNA: microRNA; pri-miRNA: primary miRNA; pre-miRNA: precursor miRNA.

4.2 lncRNAs and miRNAs: The ceRNA Hypothesis

One of the most discussed roles of lncRNAs is their ability to "sponge up" microRNAs (miRNAs), preventing them from shutting down gene expression. This forms the basis of the competing endogenous RNA (ceRNA) hypothesis. H19, for example, binds to let-7 family miRNAs, keeping them from repressing genes that promote cell growth (Kallen et al., 2013). Similarly, PTENP1, a lncRNA derived from a pseudogene, competes for miRNAs that target the tumor suppressor PTEN, helping to protect PTEN expression.

However, Denzler et al. (2014) challenged this model. Their experiments showed that for most lncRNAs, their expression levels are too low to soak up enough miRNAs to make a big difference, at least under normal conditions. Their work suggests that ceRNA effects may be limited to specific scenarios, such as cancer, where lncRNAs are often overexpressed. This introduces a critical caveat: while ceRNA activity is real, it may not be as widespread or as strong as originally believed.

5. Transcriptional regulation by lncRNA

5.1. Regulating Histone Modification at the Chromatin level

Epigenetics involves heritable changes in gene expression that occur without altering the DNA sequence. These changes are mediated by mechanisms such as DNA methylation, histone modification, chromatin remodeling, and non-coding RNAs. Among these, histone modifications, particularly methylation and acetylation of histone H3 lysine residues, are critical regulators of chromatin structure and gene activity. These modifications can either activate or silence gene expression depending on the specific chemical group added and the residue involved.

Recent studies have identified long non-coding RNAs (lncRNAs) as important regulators of histone modifications. These lncRNAs act either by directly recruiting histone-modifying enzymes or by serving as modular scaffolds that coordinate multiple epigenetic complexes at specific genomic loci.

For instance, lncPRESS1 has been shown to interact with SIRT6, a histone deacetylase, and inhibit its activity at H3K56 and H3K9, thereby preserving the acetylation status necessary for maintaining pluripotency in human embryonic stem cells (Jain et al., 2016). This represents a protective mechanism, where lncRNA safeguards gene expression by blocking repressive chromatin marks.

In contrast, GClnc1, which is upregulated in gastric cancer, functions as a scaffold lncRNA. It physically binds to a methyltransferase cofactor and a histone acetyltransferase, guiding them to specific chromatin regions to induce coordinated H3K4 methylation and H3 acetylation. This dual regulation promotes oncogene expression and supports tumor progression (Sun et al., 2016).

5.2 Enhancer-like activity of lncRNA genes

A related class of lncRNAs is the activating ncRNAs (ncRNA-as). These are a species of lncRNAs transcribed from independent loci, but not from enhancers. They also have a transcriptional activation function. ncRNA-as specifically activate the transcription of neighbouring coding genes in an RNA-dependent fashion, requiring the activity of the coding gene promoter. Thus functionally they are highly similar to eRNAs.

However, in contrast to eRNAs, ncRNA-as are spliced, polyadenylated stable transcripts. In common with both DNA enhancers and eRNAs, gene activation mediated by the ncRNA-a requires a change in chromosomal conformation to bring the ncRNA-a locus close to the promoter of its target gene. Mediator is a protein complex which, along with cohesin, is involved in bridging together enhancers and promoters.

A number of lncRNAs have been shown to be associated with the mediator complex, and depletion of the complex inhibits looping between the ncRNA-a locus and its target gene. Thus eRNA and ncRNA-a may function by interacting with the same set of molecules, forming a scaffold for a protein complex that bridges the enhancer-like element and the promoter of a coding gene (Figure 2).

While most circRNAs are cytoplasmic and thus may function in post-transcriptional regulation, a specific subclass of circRNAs demonstrates nuclear localisation and appears to function in transcriptional regulation. These circRNAs, in contrast to those described above, retain both exons and introns and therefore have been named exon–intron circRNAs (EIcircRNAs). EIcircRNAs are able to enhance transcription of their parental (coding) genes by interacting with RNAPII in an interaction that requires the U1 small nuclear RNA (snRNA).

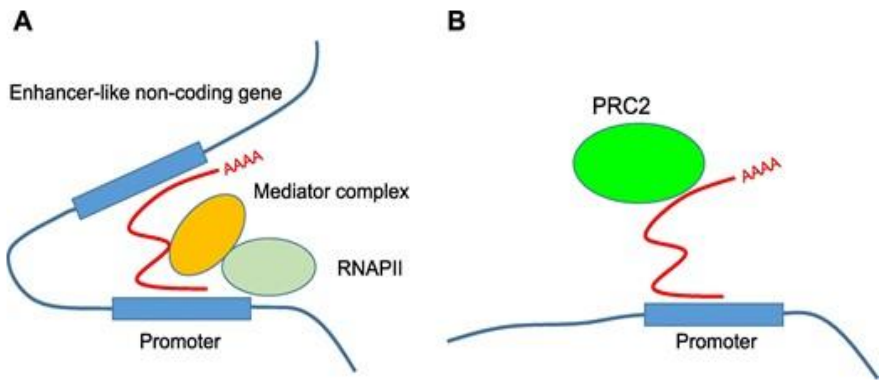

Figure 2: Models of transcriptional regulation In the bridging scaffold model.
(A), activating RNAs (red line) are transcribed from enhancer-like non-coding genes and are required to recruit the mediator complex and to mediate chromatin conformational changes bridging the enhancer-like non-coding gene and the promoter of a coding gene. (B), lncRNA (red line) recognises specific DNA motifs and recruits histone modifying enzymes such as PRC2 to the locus. lncRNA, long non-coding RNA; PRC2, polycomb repressive complex 2; RNAPII, RNA polymerase II.

6 Future directions and Current Gaps:

Research on long non-coding RNAs (lncRNAs) is growing fast, revealing their big role in controlling how genes work. Scientists are now moving beyond just listing these RNAs, they're digging into how lncRNAs interact with DNA, proteins, and other RNAs to affect gene expression. New tools like CRISPR, single-cell RNA sequencing, and advanced RNA-protein mapping are helping us understand these interactions more clearly.

Still, many lncRNAs remain a mystery. It's hard to tell whether it's the RNA itself or just the act of making it that causes changes in the cell. Also, because lncRNAs often don't have similar sequences across different species, it's tough to predict their functions or understand how they're conserved.

Some popular ideas, like lncRNAs soaking up microRNAs (the ceRNA hypothesis) or helping recruit specific protein complexes like PRC2, don't always hold true in every case. This means we need to look at each situation more carefully, especially in diseases like cancer.

Moving forward, combining genetics, cell biology, and structural studies will be key. Understanding how lncRNAs work in specific conditions could open the door to new treatments and tools for diagnosing diseases.

7 Conclusion:

This review underscores the crucial roles of long non-coding RNAs (lncRNAs) in regulating gene expression at both transcriptional and post-transcriptional levels. LncRNAs are no longer viewed as transcriptional noise but as functional molecules that influence chromatin remodeling, transcription factor activity, enhancer–promoter looping, and histone modifications. Post-transcriptionally, they regulate mRNA splicing, stability, and translation, and also act as precursors or sponges for microRNAs, adding another layer of gene expression control.

Despite these advances, several key questions remain unresolved. Many lncRNAs lack clear functional annotation, and distinguishing the effects of lncRNA transcripts from those of their transcriptional loci remains a challenge. Furthermore, mechanisms such as PRC2 recruitment and ceRNA activity are still debated due to variability across biological contexts. The limited sequence conservation of lncRNAs also raises questions about how their functions are preserved across species.

To address these gaps, future research should employ integrative, high-resolution approaches such as CRISPR-based editing and advanced RNA-protein interaction mapping. Understanding the specific roles of lncRNAs in different tissues and disease states is essential for realizing their full potential.

In conclusion, lncRNAs represent a complex and versatile component of gene regulation, with promising implications for both basic biology and clinical applications.

References used:

1. Engreitz, J. M. et al. (2016) Local regulation of gene expression by lncRNA promoters, transcription and splicing. Nature, 539(7629), 452–455.

2. da Rocha, S.T. · Heard, E. Novel players in X inactivation: insights into Xist-mediated gene silencing and chromosome conformation. Nat. Struct. Mol. Biol. 2017; 24:197204.

3. Chu, C. · Zhang, Q.C. · da Rocha, S.T. Systematic discovery of Xist RNA binding proteins Cell. 2015; 161:404-416.

4. McHugh, C.A. · Chen, C.K. · Chow, A. The Xist lncRNA interacts directly with SHARP to silence transcription through HDAC3 Nature. 2015; 521:232-236.

5. Minajigi, A. · Froberg, J. · Wei, C. Chromosomes. A comprehensive Xist interactome reveals cohesin repulsion and an RNA-directed chromosome conformation Science. 2015; 349.

6. Monfort, A. · Di Minin, G. · Postlmayr, A. Identification of Spen as a Crucial Factor for Xist Function through Forward Genetic Screening in Haploid Embryonic Stem Cells Cell Rep. 2015; 12:554-561.

7. Anderson, K.M. · Anderson, D.M. · McAnally, J.R. Transcription of the non-coding RNA upperhand controls Hand2 expression and heart development, Nature. 2016; 539:433-436.

8. Latos, P.A. · Pauler, F.M. · Koerner, M.V. Airn transcriptional overlap, but not its lncRNA products, induces imprinted Igf2r silencing, Science. 2012; 338:1469-1472.

9. Davidovich, C. · Wang, X. · Cifuentes-Rojas, C. Toward a consensus on the binding specificity and promiscuity of PRC2 for RNA, Mol. Cell. 2015; 57:552-558.

10. Petruk, S. · Sedkov, Y. · Riley, K.M. Transcription of bxd noncoding RNAs promoted by trithorax represses Ubx in cis by transcriptional interference, Cell. 2006; 127:1209-1221.

11. Rinn, J.L. · Kertesz, M. · Wang, J.K. Functional demarcation of active and silent chromatin domains in human HOX loci by noncoding RNAs, Cell. 2007; 129:1311-1323.

12. Portoso, M. · Ragazzini, R. · Brenčič, Ž. PRC2 is dispensable for HOTAIR-mediated transcriptional repression, EMBO J. 2017; 36:981-994.